

PA6.6 – Polyamide 6.6 PA66 + PA6I/6T GF50

AKROLOY® PA GF 50 black (6507)

Tensile modulus

17000 MPa

1 mm/min

ISO 527-2

Stress at break

265 MPa

5 mm/min

ISO 527-2

Charpy impact strength

105 kJ/m²

23°C

ISO 179-1/1eU

AKROLOY® PA GF 50 black (6507) is a 50% glass fibre reinforced, semi-aromatic polyamide-blend with very high stiffness and strength, even in conditioned state.

Typical applications

Components with high dimensional stability, independent from moisture content. AKROLOY PA GF 50 black is an alternative for aluminium- and zinc diecast alloys.

**Mechanical Properties**

Tensile modulus (1 mm/min | ISO 527-2)

d.a.m.

17000 MPa

conditioned

15500 MPa

Stress at break (5 mm/min | ISO 527-2)

d.a.m.

265 MPa

conditioned

190 MPa

Strain at break (5 mm/min | ISO 527-2)

d.a.m.

2,7 %

conditioned

2,7 %

Flexural modulus (2 mm/min | ISO 178)

d.a.m.

16500 MPa

conditioned

15000 MPa

Flexural strength (2 mm/min | ISO 178)

d.a.m.

385 MPa

conditioned

285 MPa

Flexural strain at break (2 mm/min | ISO 178)

d.a.m.

2,8 %

conditioned

2,6 %

Charpy impact strength (23°C | ISO 179-1/1eU)

d.a.m.

105 kJ/m²

conditioned

100 kJ/m²

Charpy impact strength (-30°C | ISO 179-1/1eU)

d.a.m.

95 kJ/m²

Charpy impact strength (-40°C | ISO 179-1/1eU)

d.a.m.

95 kJ/m²

Charpy notched impact strength (23°C | ISO 179-1/1eA)

d.a.m.

18 kJ/m²

conditioned

17 kJ/m²

Charpy notched impact strength (-30°C | ISO 179-1/1eA)

d.a.m.

17 kJ/m²

Charpy notched impact strength (-40°C | ISO 179-1/1eA)

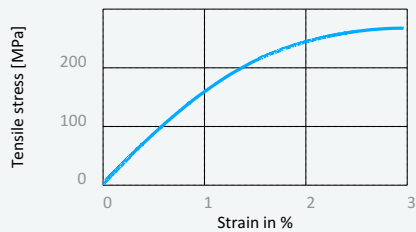
d.a.m.

17 kJ/m²

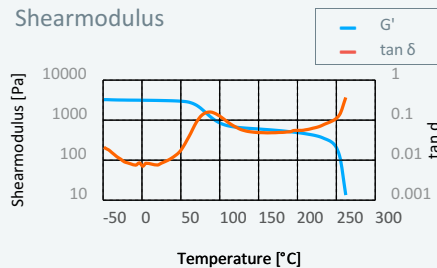
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Stress strain chart at 23°C



Shear modulus

**Thermal Properties**

Temperature of deflection under load HDT/A (1,8 MPa ISO 75)	245 °C
Temperature of deflection under load HDT/C (8 MPa ISO 75)	185 °C
Melting temperature (DSC, 10K/min DIN EN ISO 11357-3)	255 °C
Coefficient of linear thermal expansion, parallel (23°C bis 80°C ISO 11359-1/2)	0,15 1,0E-4/K
Coefficient of linear thermal expansion, transverse (23°C bis 80°C ISO 11359-1/2)	0,65 1,0E-4/K
Temperature index for 50% loss of tensile strength after 5.000h (5.000 h IEC 60216)	150 °C
Temperature index for 50% loss of tensile strength after 20.000h (20.000 Std. IEC 60216)	115 °C

**Flammability**

Burning rate (UL 94) 1,6mm Wall thickness	HB Class
GWFI (IEC 60695-2-12) 1,6mm Wall thickness	650 °C
GWIT (IEC 60695-2-13) 1,6mm Wall thickness	675 °C
Burning rate (<100 mm/min) (> 1 mm Thickness FMVSS 302)	+

**General properties**

Density (23°C ISO 1183)	1,56 g/cm³
Humidity absorption (70°C, 62% r.H. ISO 1110)	1,3-1,5 %
Molding shrinkage (flow ISO 294-4)	0,1-0,3 %
Molding shrinkage (transverse ISO 294-4)	0,3-0,5 %

**Rheological Properties**

Flowability (1mm Thickness AKRO)	100 mm
Flowability (2mm Thickness AKRO)	260 mm

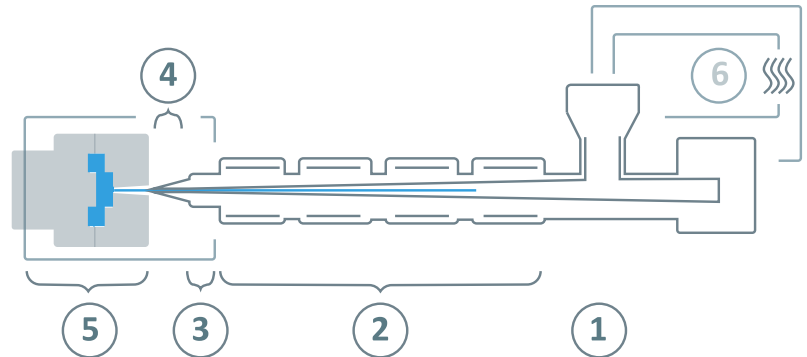
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AKROLOY® PA GF 50 black (6507)**Processing information**

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only dehumidifying or vacuum dryers. Extensive drying can cause filling problems and surface defects.



⑥	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80°C
	Processing moisture	0,02 - 0,1%
①	Feed section	60 - 80°C
②	Temperature zone 1 - Zone 4	260 - 300°C
③	Nozzle temperature	270 - 300°C
④	Melt temperature	280 - 300°C
⑤	Mold temperature	90 - 130°C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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